

* Deep Learning

Machine learning framework fail to work with images, that's why we introduce deep learning

Deep learning: collection of neural network

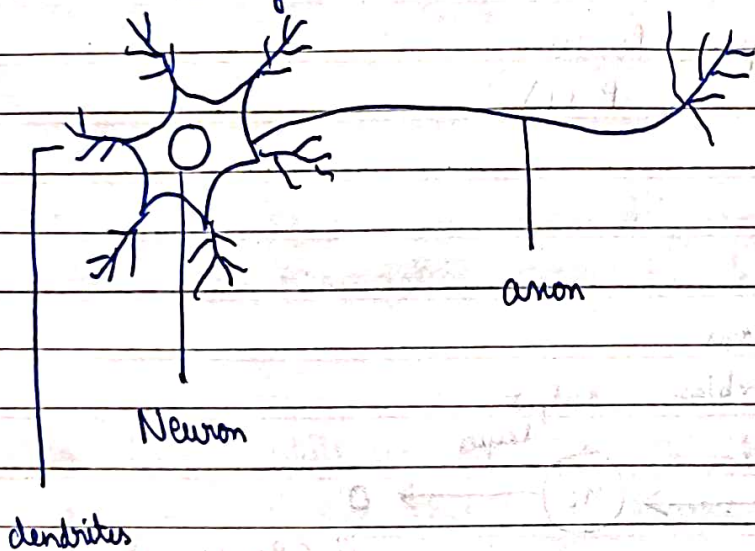
ANN (Artificial Neural N/w)

CNN (Convolution neural N/w)

RNN (Recurrent neural n/w)

In our brain one neuron is connected with another neuron and forms a neural network.

Neural working

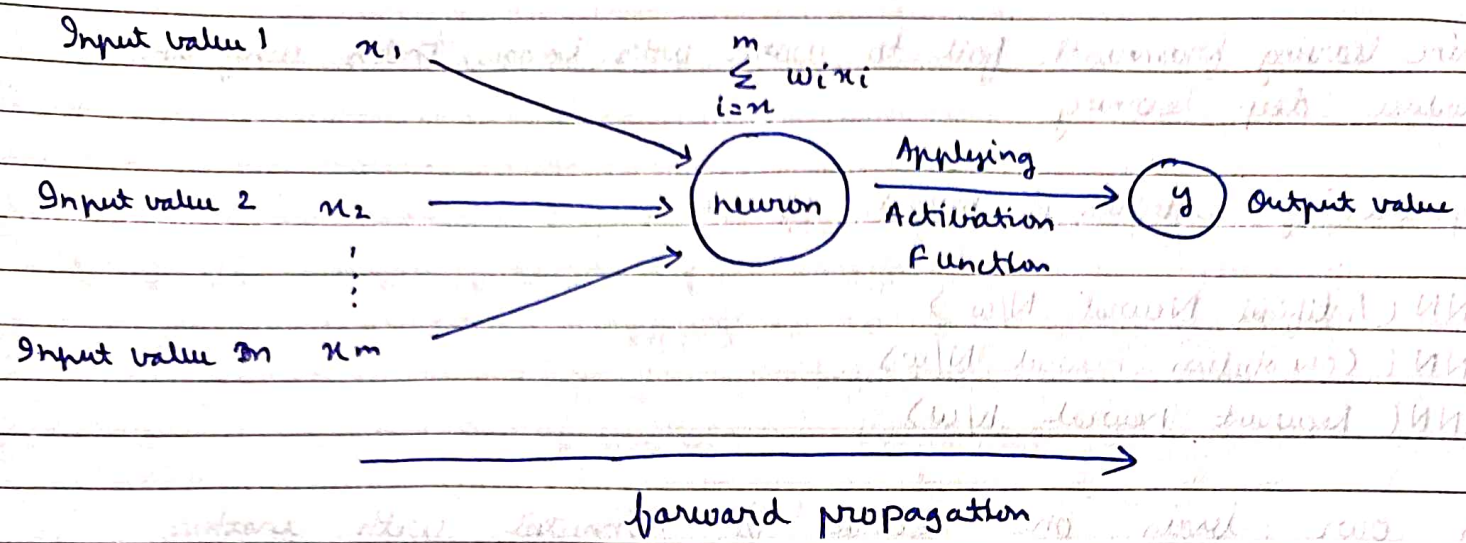


① Single layer neural network

1 input layer - 1 hidden layer - 1 output layer

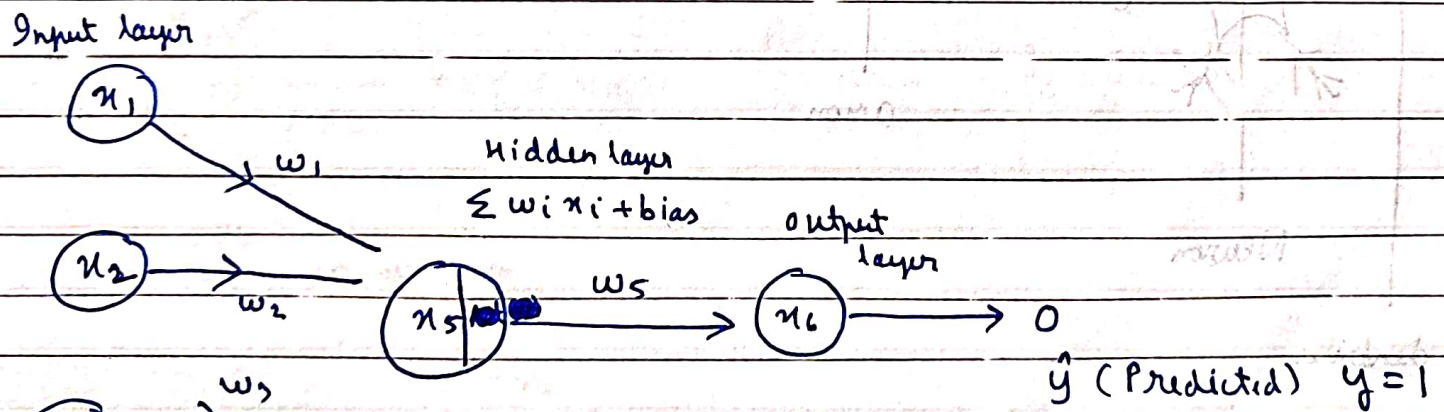
② Multi layer neural network

1 input layer - multiple hidden layer - 1 output layer.



g:

x_1	x_2	x_3	x_4	Result
Study	Play	eat	sleep	P(1)
5	2	1	8	



Forward propag

$$\text{Loss} = \text{Actual} - \text{Predicted}$$

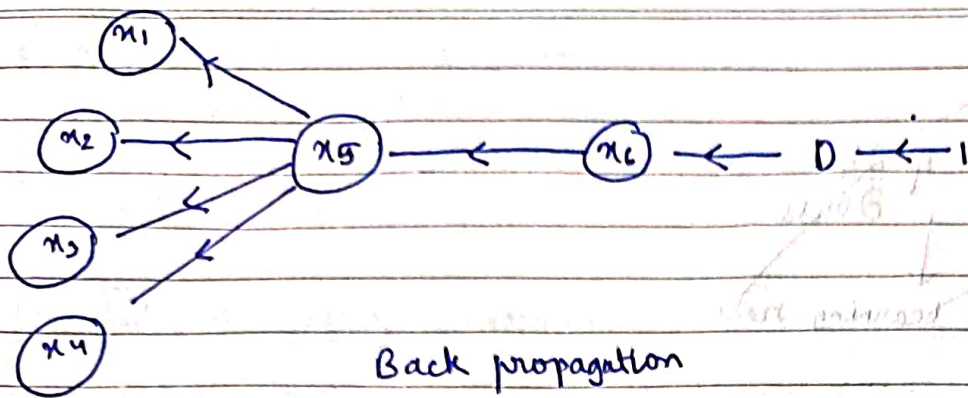
$$= y - \hat{y}$$

$$= 1 - 0$$

$$= 1$$



This should be near equal to 0.



$$1 \text{ FP} + 1 \text{ BP} = 1 \text{ iteration} = \text{EPOCH}$$

fine-tune of weights -- weights are changed every back propagation while we perform epoch.
 when fine tuning of weights will happen model will predict good result (Optimized result)

Hence now to reduce this difference we need to update the weights while back propagation with the help of optimizers.

Sometimes while we do the epoch the new weight = old weight the iterⁿ stops. This is called vanishing gradient descend.
 To solve this problem delete a neuron or perform regularization.

* Activation Function

- ① Sigmoid ② Tanh ③ ReLU ④ LeakyReLU ⑤ Softmax

① Sigmoid activation fn is always implemented on the output layer.
Range of sigmoid curve -- 0 to 1.

② tanh

If the data contains -ve value then sigmoid function will fail

tanh range is -1 to 1

tanh activation function implement in output layer.

③ ReLU

ReLU activation function $\max(x, 0)$

$\max(0, -1) = 0$

$\max(10, 100) = 100$

ReLU never gives any -ve value

max -- maximum value only pass from one neuron to another neuron.

ReLU will implement in Hidden layer.

④ LeakyReLU

If the data has -ve value range so we introduce another activation function.

Leaky ReLU range -1 to ∞

⑤ Softmax (multiclass classification)

probability distribution

range will be 0-1.

output layer

* Convolutional neural network

Input data is images, video, live camera.

Image processing is a method to perform some operations on an image so as to get an enhanced image or to extract some info from an image.

computer vision - cv2 is a library framework which is helpful for image recognition, image classification.

image matrix of pixel.

2d channel -- black and white

3d channel -- BGR

Every pixel ranging b/w 0-255.

light side color $\Rightarrow$ 0-50

dark side color $\Rightarrow$ 200-250

Every image is broken down into pixels which are further divided into numbers.

CNN

- Step 1: Convolution
- Step 2: Max Pooling
- Step 3: Flattening
- Step 4: Full Connection

* ANN v/s CNN

ANN

CNN

Input layer

Input matrix

Hidden layer

filter or feature detector

Output layer

output matrix or feature map

Rest of the things are same in both cases.

① Convolution layer = math operation between input matrix $\times$ filter = output matrix

Two main concepts we have in this case striding and padding.

② After output layer we will be applying max pooling.

③ Flattening is used to convert all the resultant 2-D arrays from pooled feature maps to a single long continuous linear vector.